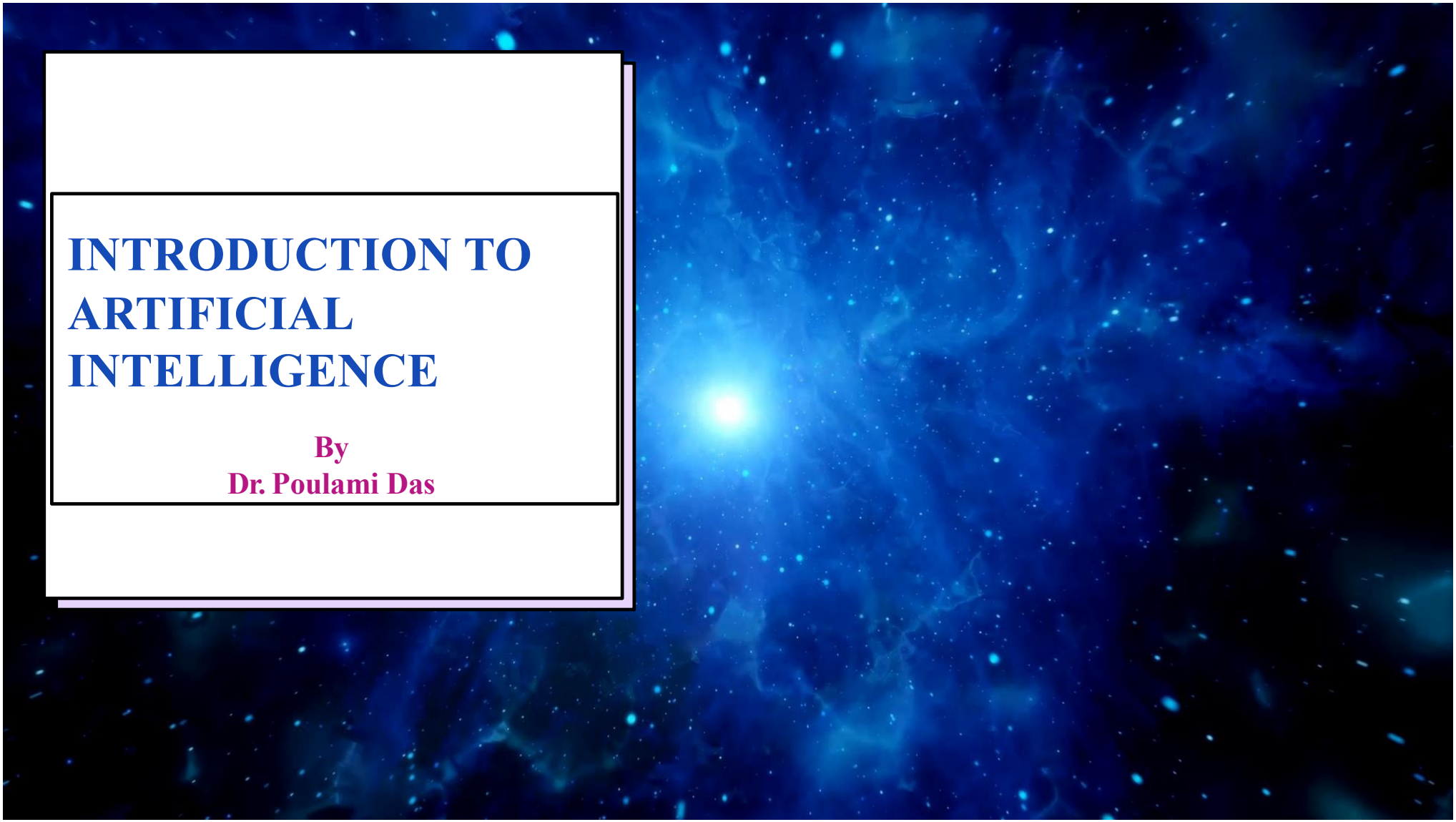

INTRODUCTION TO
ARTIFICIAL
INTELLIGENCE
By
Dr. Poulami Das

Goals of the Course

- A brief intro to the philosophy of AI
- A brief intro to the breadth of ideas in AI

- General computer scientist
  - general tools to aid in attacking a new problem

- Serious AI enthusiast
  - A primer from which to launch advanced study



Some Fundamental Questions

What is intelligence?

What is thinking?

What is a machine?

Is the computer a machine?

Here on when we say machine we will mean a programmable computer system

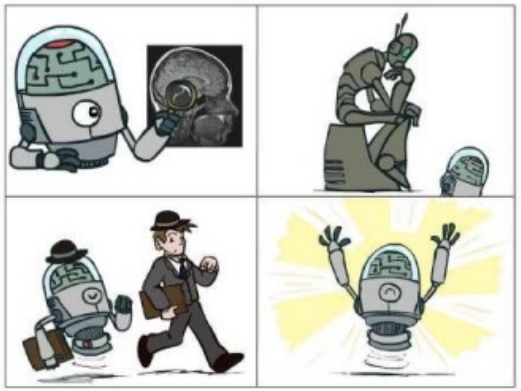
Can a machine think ?

If yes are we machines?!



What is AI?

Think like people



Think rationally

Act like people

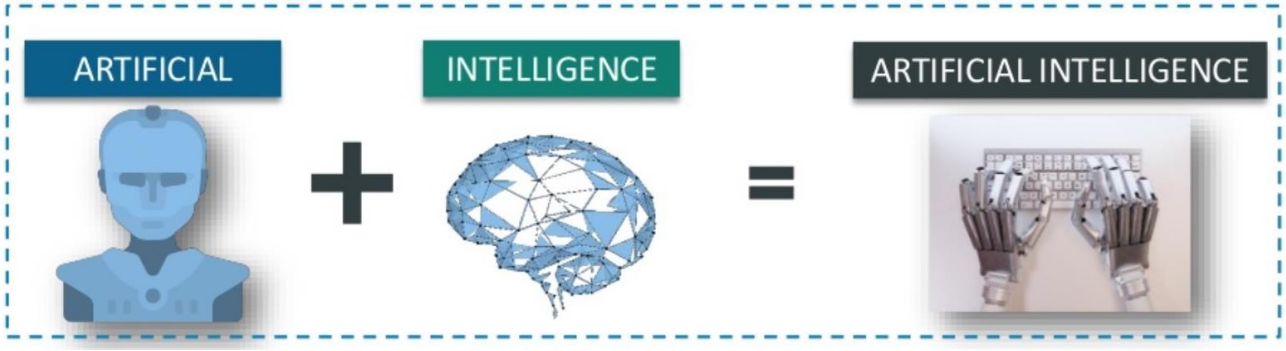
Act rationally

Thinking humanly: The automation of activities that we associate with human thinking, activities such as decision-making, problem solving, learning...(Richard Bellman, 1978)

Acting Humanly: "The art of creating machines that perform functions that require intelligence when performed by people." (Ray Kurzweil, 1990)

Acting rationally: AI . . . is concerned with intelligent behavior in artifacts (Nilsson, 1998)

Artificial Intelligence - Definition

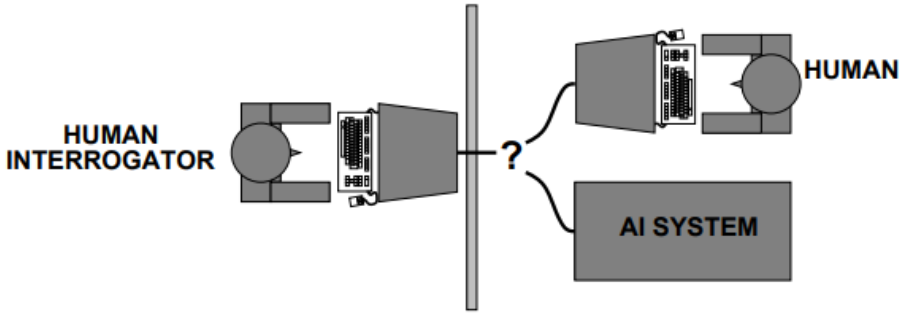


- Artificial Intelligence is an area of Computer Science that emphasizes the creation of intelligent machines that work and react like humans.
- Artificial Intelligence is intelligence exhibited by machines, rather than the humans or animals.

Acting humanly: The Turing test

Turing (1950) "Computing machinery and intelligence":

- "Can machines think?" $\longrightarrow$ "Can machines behave intelligently?"
- Operational test for intelligent behavior: the Imitation Game



- Predicted that by 2000, a machine might have a 30% chance of fooling a lay person for 5 minutes
- Suggested major components of AI: knowledge, reasoning, language understanding, learning
- Total Turing test: computer vision, robotics

Thinking humanly: Cognitive Science

1960s "cognitive revolution": information-processing psychology replaced prevailing orthodoxy of behaviorism

Get inside: introspection, psychological experiments, brain imaging

Requires scientific theories of internal activities of the brain
- What level of abstraction? "Knowledge" or "circuits"?
- How to validate? Requires
  1) Predicting and testing behavior of human subjects (top-down)
  or 2) Direct identification from neurological data (bottom-up)

Both approaches (roughly, Cognitive Science and Cognitive Neuroscience) are now distinct from AI

Both share with AI the following characteristic: **the available theories do not explain anything resembling human-level general intelligence**. Hence, all three fields share one principal direction!





Thinking rationally: Laws of Thought

Normative (or prescriptive) rather than descriptive

Aristotle: what are correct arguments/thought processes?

Several Greek schools developed various forms of logic:
notation and **rules of derivation** for thoughts;
may or may not have proceeded to the idea of mechanization

Logicist tradition withi AI hopes to build on such programs to create intelligent systems.

Problems:
1) Not all intelligent behavior is mediated by logical deliberation
2) What is the purpose of thinking? What thoughts **should** I have
out of all the thoughts (logical or otherwise) that I **could** have?



Acting rationally

Rational behavior: doing the right thing

The right thing: that which is expected to maximize goal achievement, given the available information

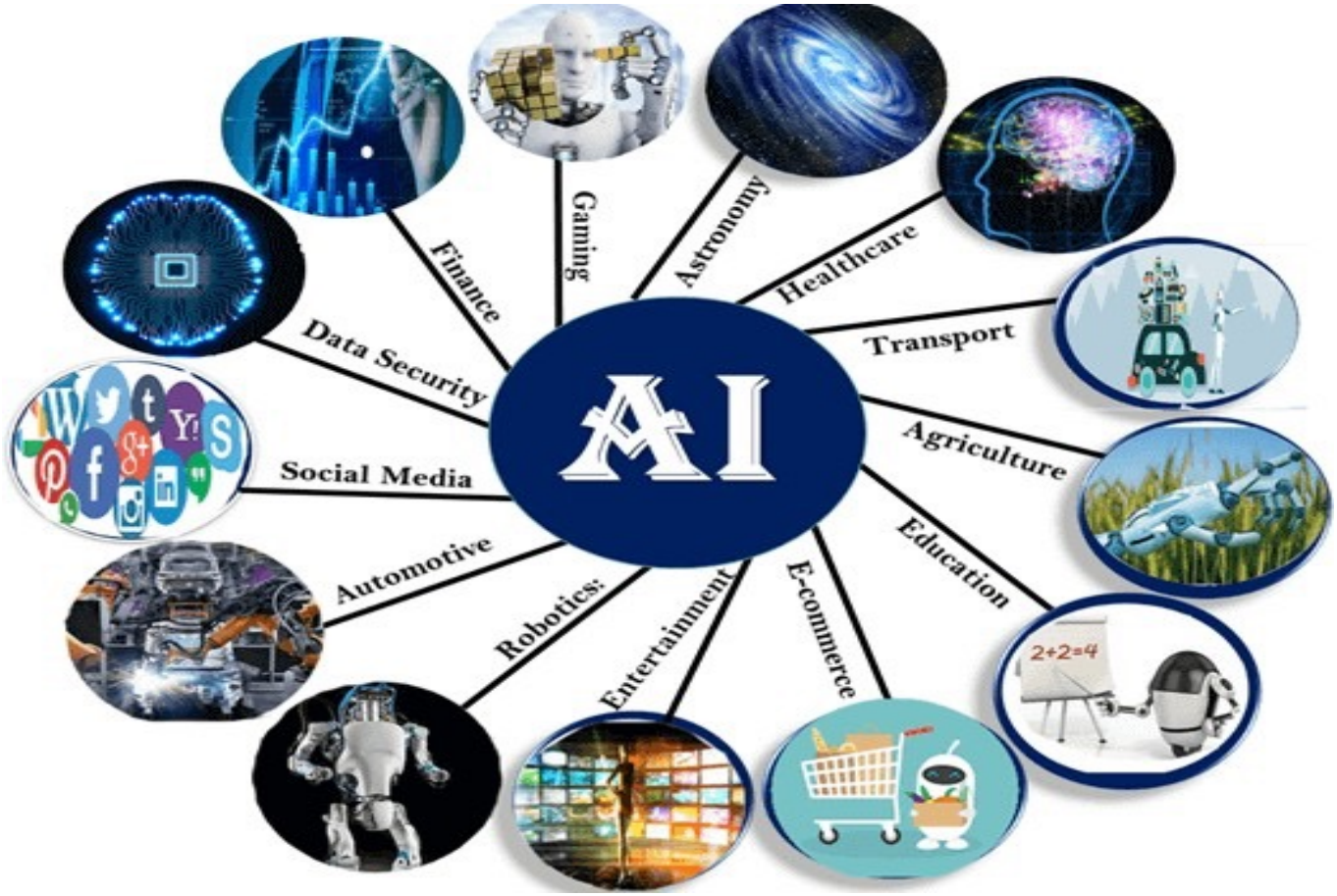
Doesn't necessarily involve thinking—e.g., blinking reflex—but thinking should be in the service of rational action

Aristotle (Nicomachean Ethics):

> **Every art and every inquiry, and similarly every action and pursuit, is thought to aim at some good**



Applications of Artificial Intelligence

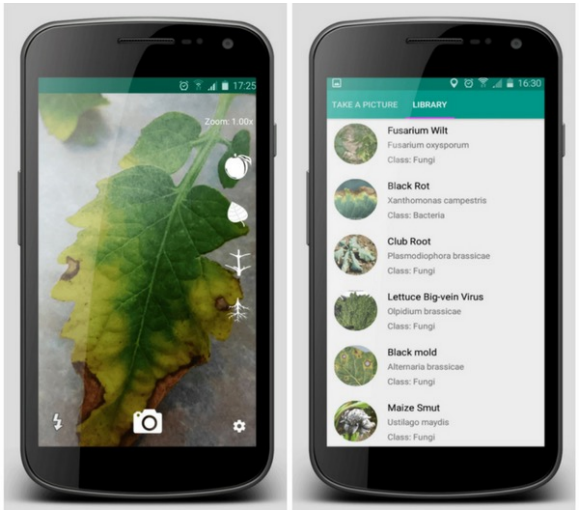


In Agriculture

A firm called Blue River Technology has created a robot known as See & Spray. It uses object detection like computer vision technologies to control and spray weedicide on cotton plants.

Another firm PEAT is developing an application known as Plantix which checks defects and nutrient deficiencies in the soil and abnormalities in plants using images.







In Gaming

There is a tremendous application of AI in the gaming industry. There is a firm called DeepMind, which has developed a game that defeated Chess champ Garry Kasparov.
Another example is the First Encounter Assault Recon, which is a first person shooter video game.

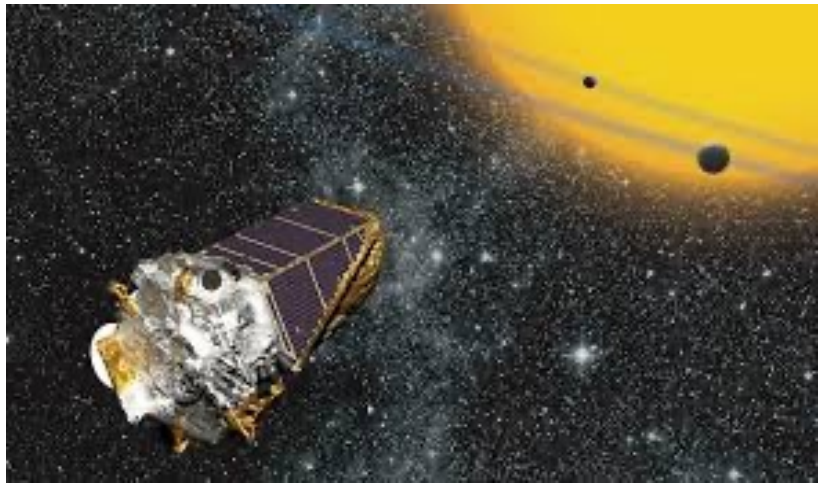
"I always say I was the first knowledge worker whose job was threatened by a machine" – Garry Kasparov

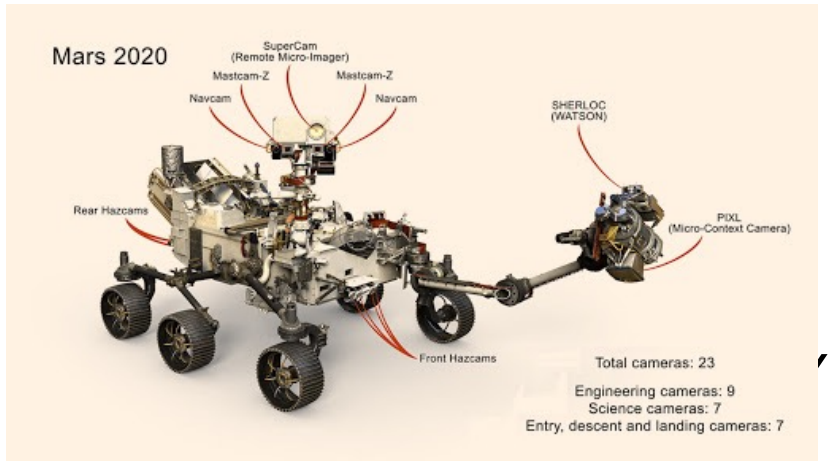




In Space Exploration

AI is used extensively in space exploration. Space exploration is used to analyze lots of data.
Astronomers used AI to sift through lots of data from the Kepler telescope for identifying a solar planet system.
Another instance of AI being used is Mars 2020 Rover. It is also known as AEGIS and is already on MARS. The rover is used to target cameras for performing investigations on MARS.





In Self-driving vehicles

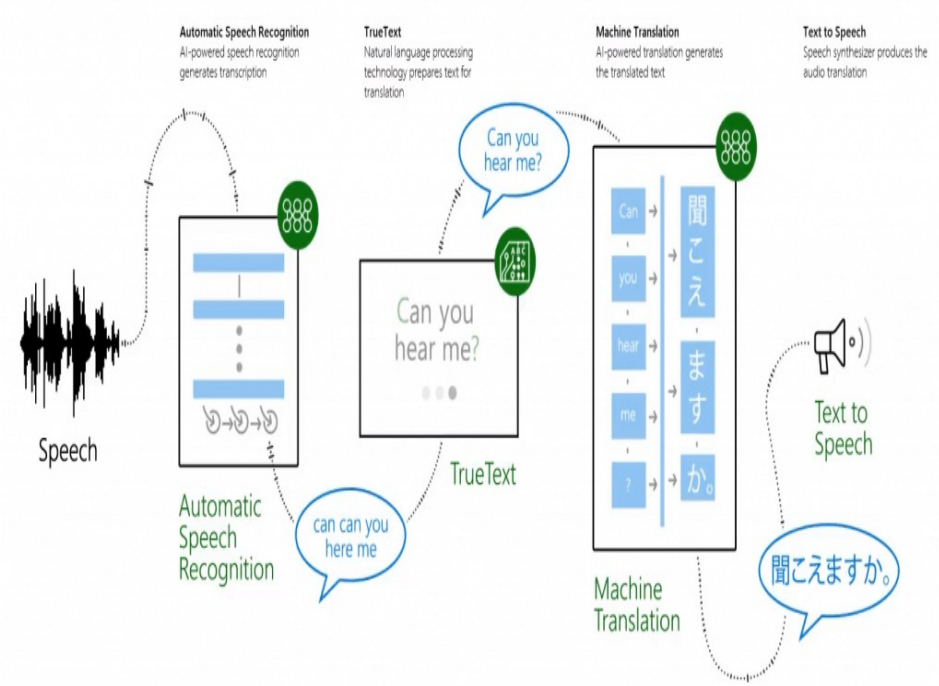
AI is instrumental in the case of self driving vehicles. Firms like Waymo are deploying their AI based public ride hailing service. The AI system collects data from cameras, GPS, vehicles radar, and cloud services to produce control signals which operate the vehicle.

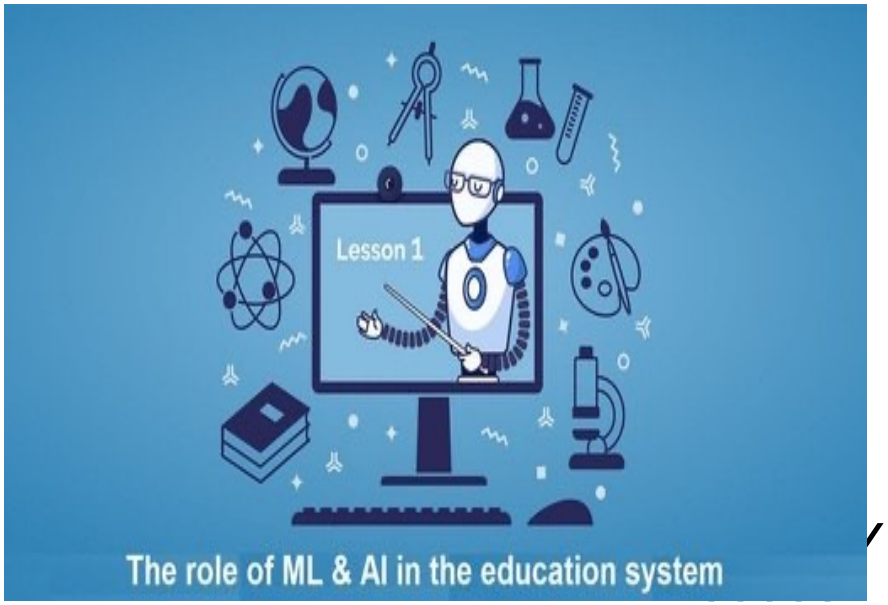
Another example is Tesla's self driving car. AI is used to implement image detection, computer vision and deep learning for making cars which automatically drive and check objects.



In Education

Teacher and AI collaboration.
Differentiated and individualized learning.
Universal access for all students.
Support outside the classroom.

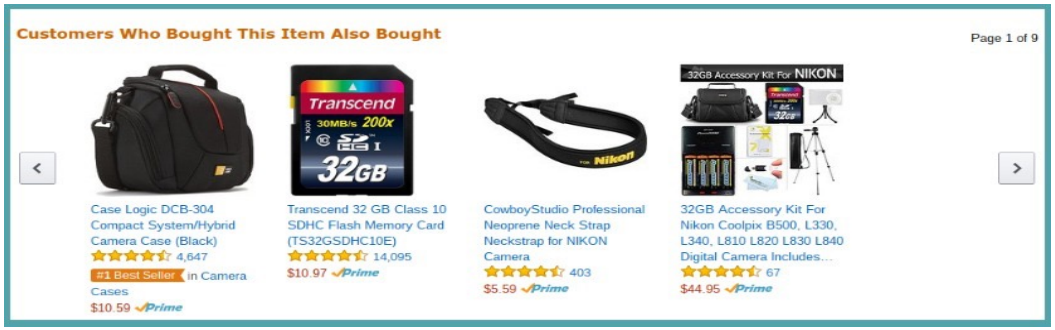


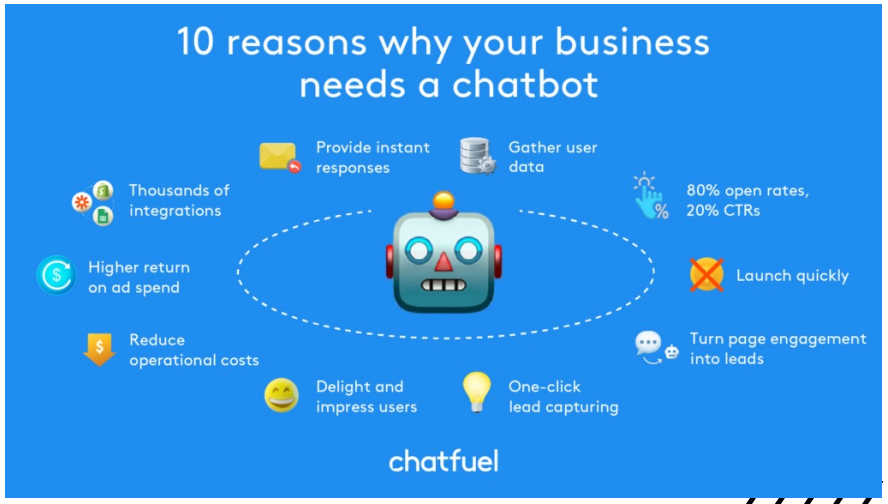


In Business

- Chatbots, virtual assistants, and business intelligence bots
- Targeted online advertising
- Predictive analytics
- Voice recognition
- Pattern recognition using Clustering
- Recommendation using Clustering

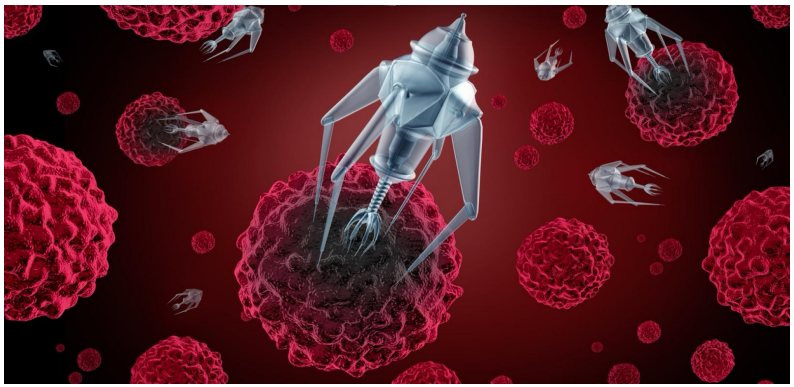




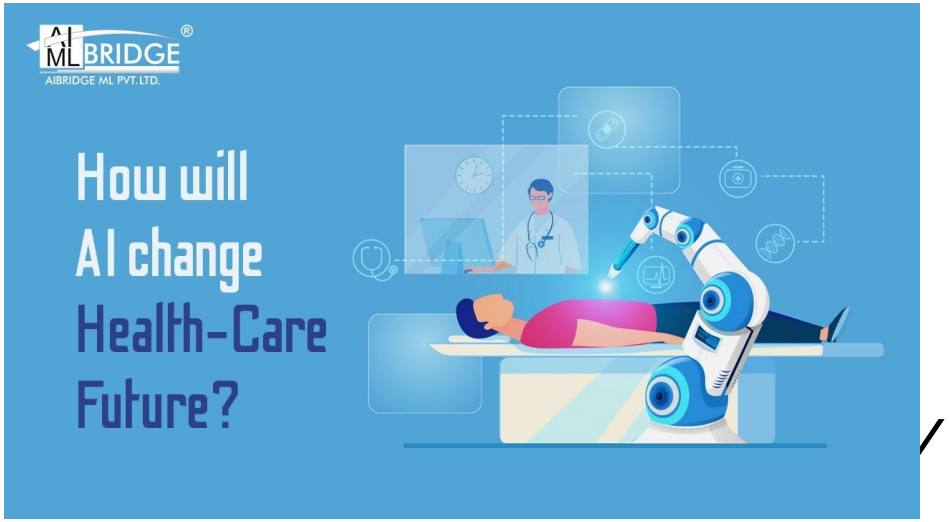


In Healthcare





Telemedicine.
Nanorobots for operations.
Medical chatbots.
Image Processing for detection of diseases.
Medical data analysis for prediction.
Automated prescription reading app.

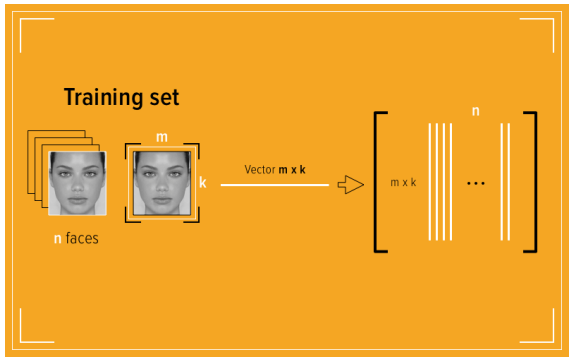


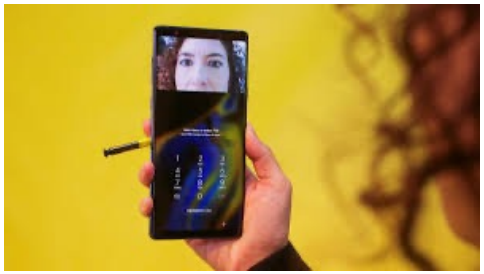
In Everyday Life













Smart homes, Smart hotels, Smart Industries - Smart Cities



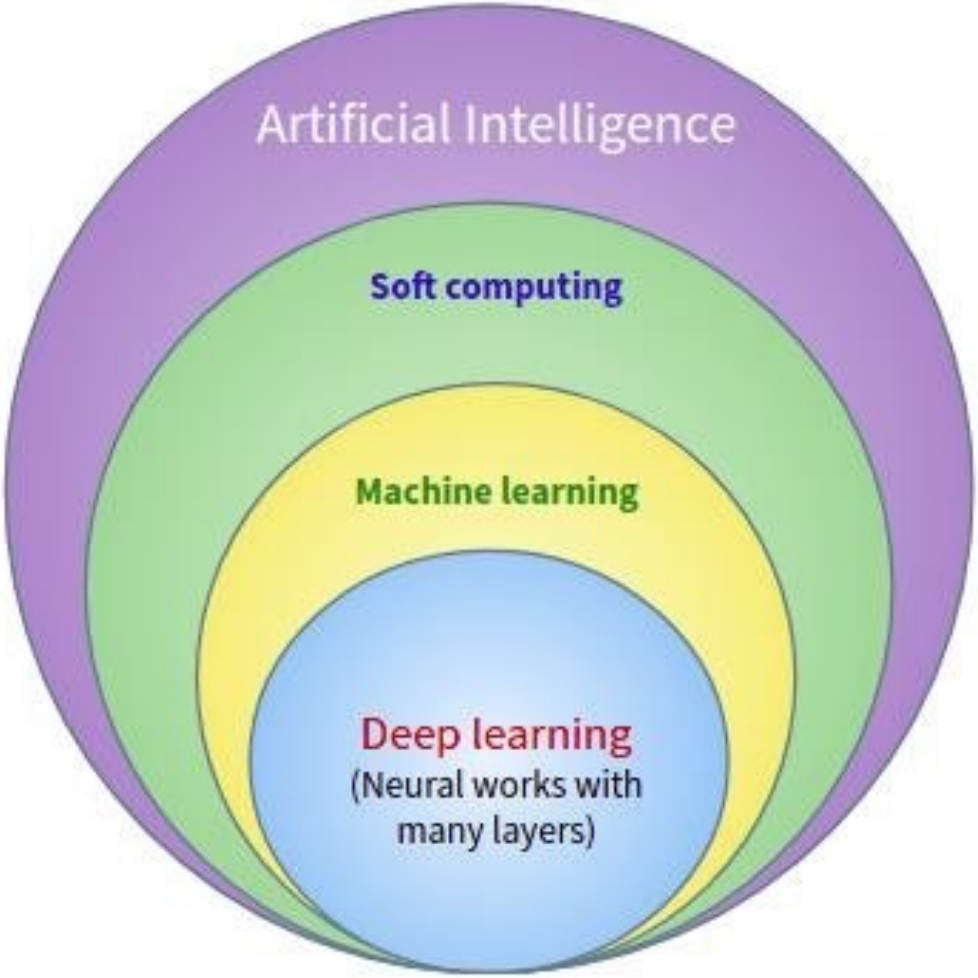


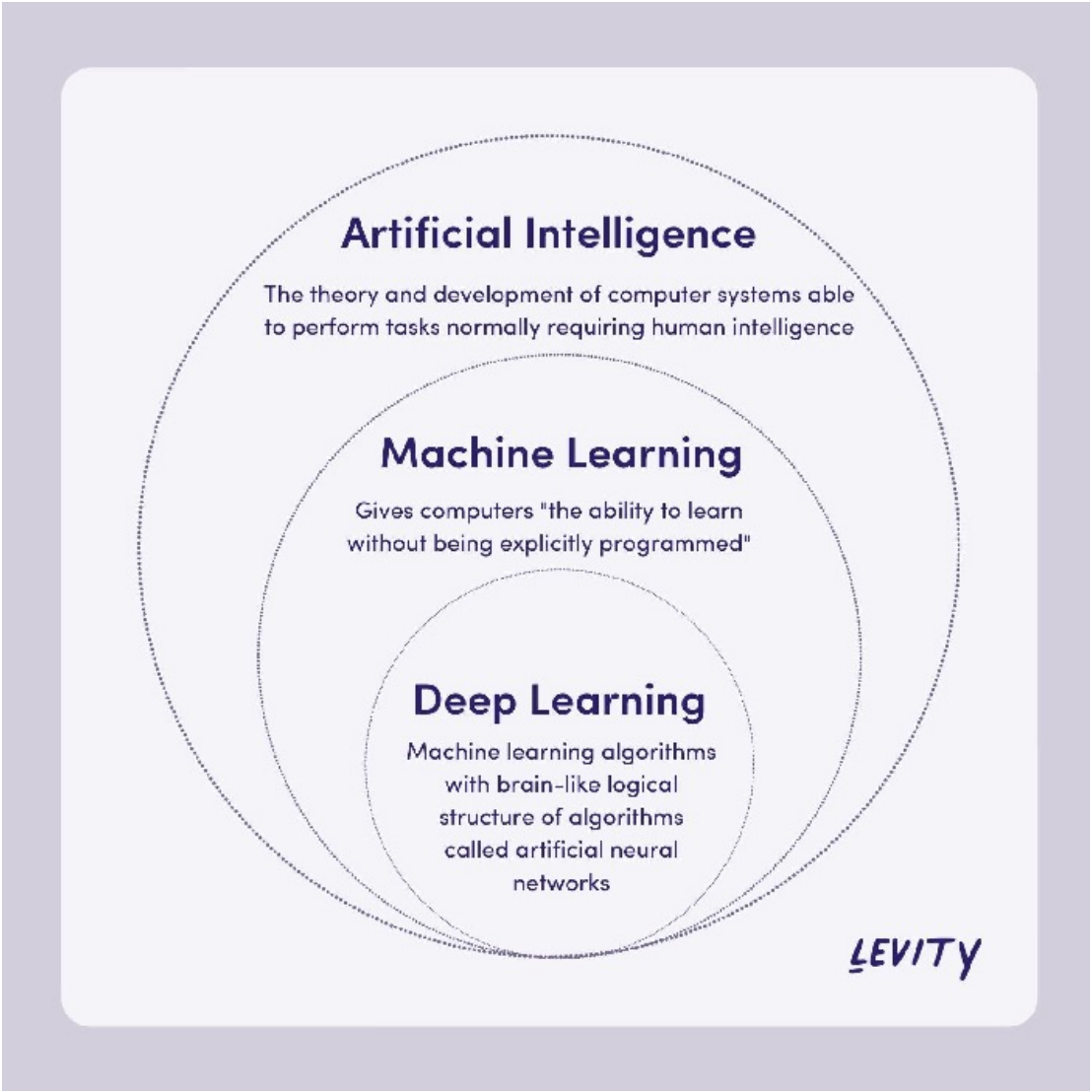




"The coming era of Artificial Intelligence will not be the era of war, but be the era of deep compassion, non-violence, and love."

- Amit Ray, Pioneer of Compassionate AI Movement


Artificial Intelligence
Soft computing
Machine learning
Deep learning
(Neural works with
many layers)


Artificial Intelligence
The theory and development of computer systems able to perform tasks normally requiring human intelligence
Machine Learning
Gives computers "the ability to learn without being explicitly programmed"
Deep Learning
Machine learning algorithms with brain-like logical structure of algorithms called artificial neural networks
LEVITY

What is Machine Learning?

Machine Learning is a class of algorithms which is data-driven, i.e. unlike "normal" algorithms it is the data that "tells" what the "good answer" is







Getting computers to program themselves and also teaching them to make decisions using data

"Where writing software is the bottleneck, let the data do the work instead."

Features of Machine Learning



01

It uses the data to *detect patterns* in a dataset and *adjust program actions accordingly*

It *focuses on* the *development of computer programs* that can teach themselves to *grow and change* when *exposed* to *new data*

02





03

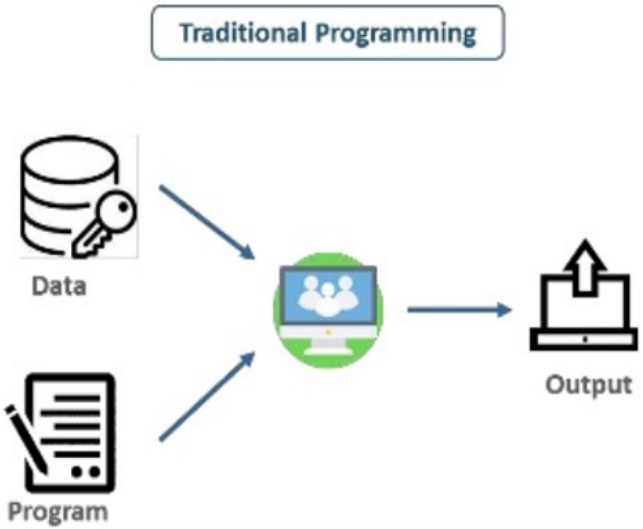
It enables computers to *find hidden insights using iterative algorithms without being explicitly programmed*

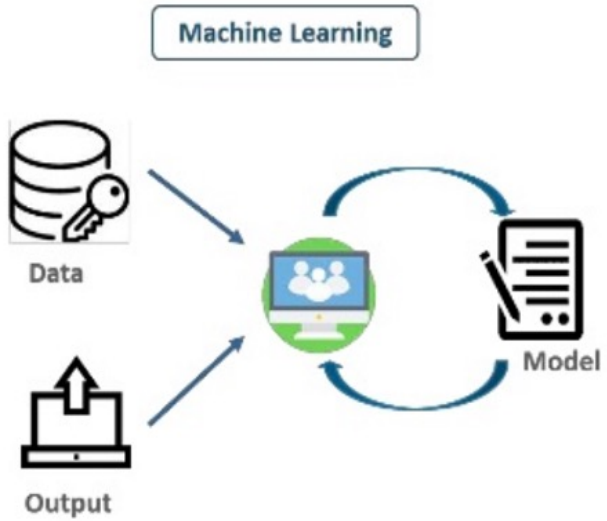
Machine learning is a *method* of *data analysis* that *automates analytical model building*

04



How It Works?

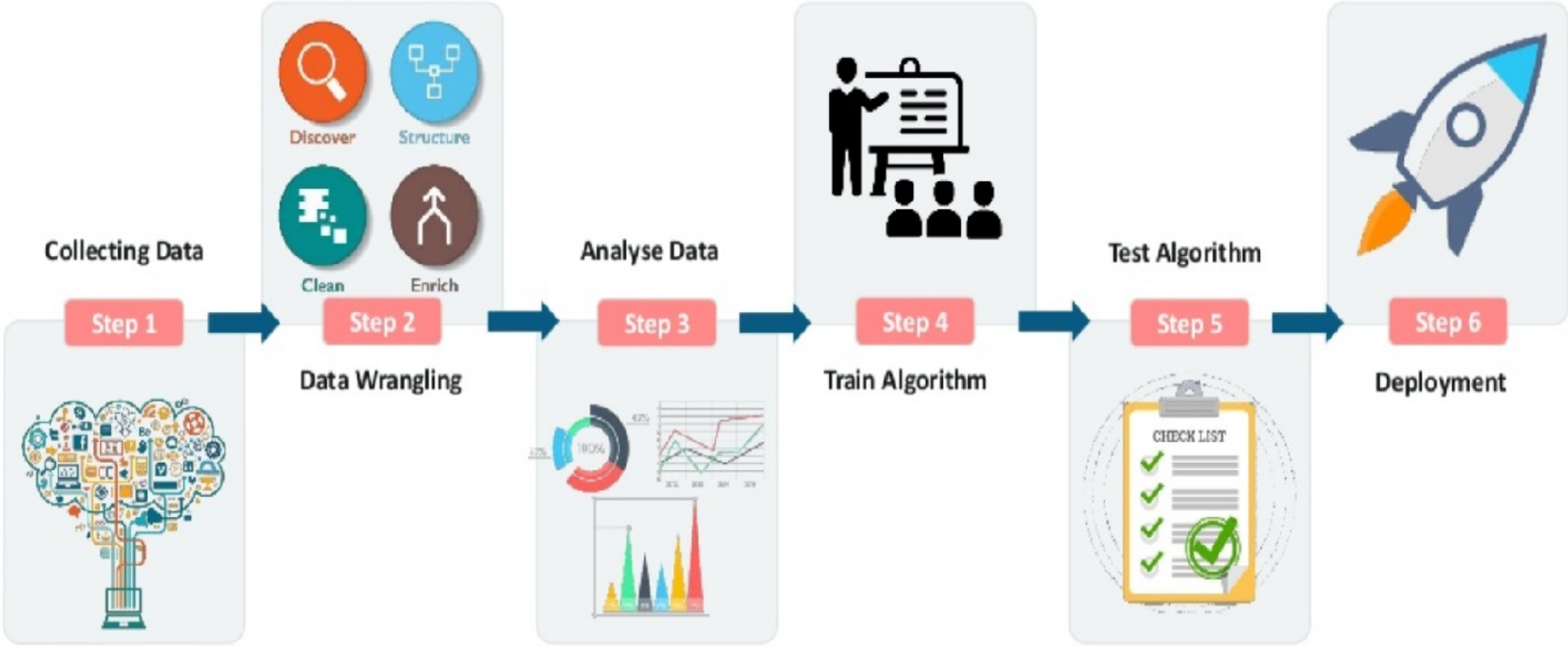


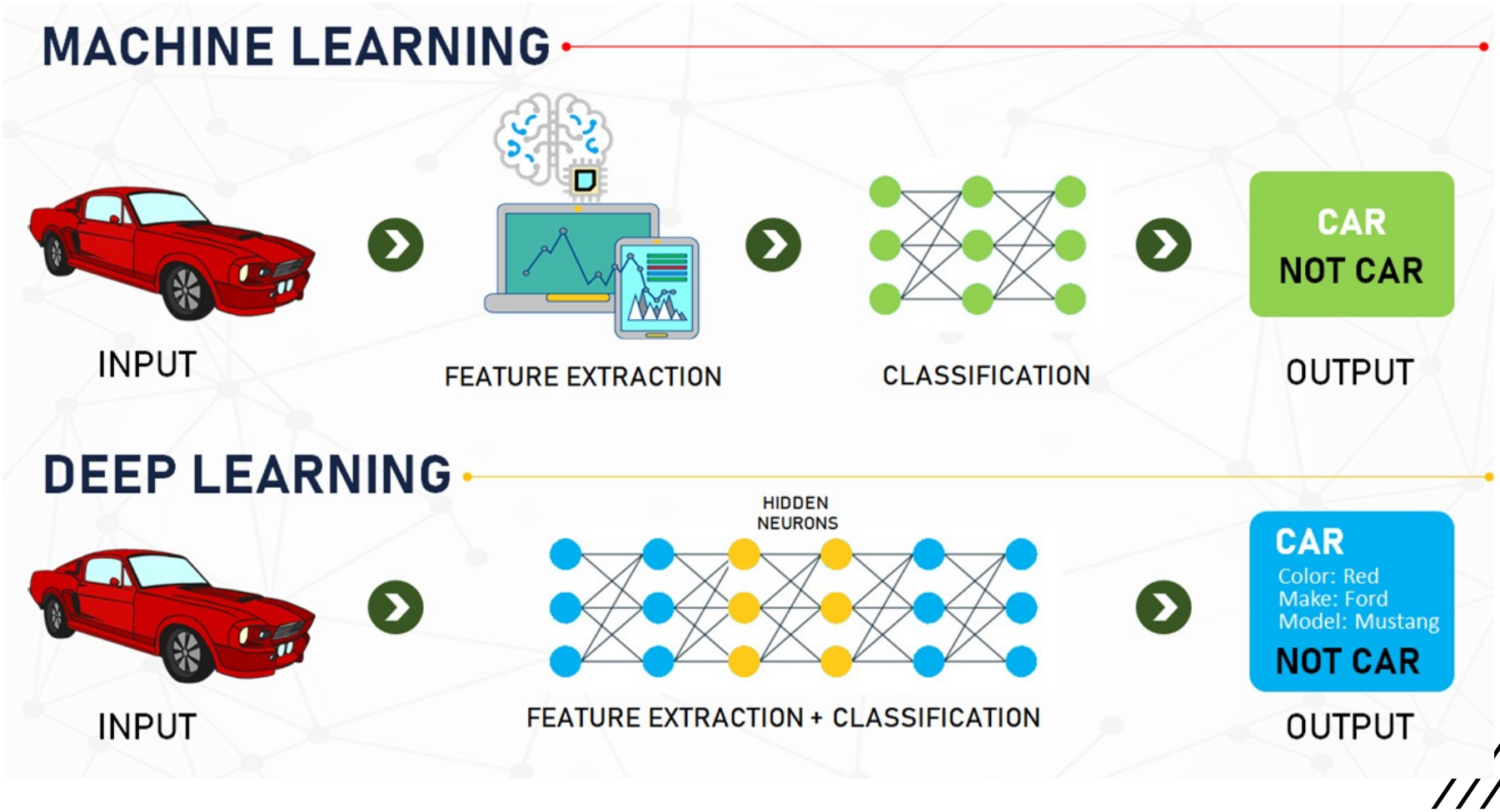




- *Learn from Data*
- *Find Hidden Insights*
- *Train and Grow*

Machine Learning Life Cycle




MACHINE LEARNING
INPUT
FEATURE EXTRACTION
CLASSIFICATION
CAR
NOT CAR
OUTPUT
DEEP LEARNING
HIDDEN
NEURONS
INPUT
FEATURE EXTRACTION + CLASSIFICATION
CAR
Color: Red
Make: Ford
Model: Mustang
NOT CAR
OUTPUT

THANK YOU